Tabular Foundation Models for Rare Classes: In-Context Learning from a Handful of Biomedical Samples

1 Context of the internship

The internship will take place at Télécom Paris, within the Image, Données, Signal (IDS) Department, and more specifically in the IMAGES research team. The internship is expected to start in spring 2027 and will have a duration of four to six months. It will be conducted in a research-oriented environment, with close interaction with academic researchers and PhD students working on machine learning for biomedical data and medical imaging.

2 Internship issue

In biomedicine, many clinically important categories are rare: rare disease subtypes, uncommon arrhythmias, rare tumour types, or melanoma among benign skin lesions. For such classes, only a handful of labelled patients may be available. Tabular foundation models (TFMs), such as TabPFN [1] and TabICL [2], predict by in-context learning: the labelled samples are given as a prompt and no training is required. They are now state-of-the-art on small tables, which makes them attractive in these settings. This internship investigates how TFMs behave when a class is represented by only a few samples within a long-tailed, multi-class problem, and how their predictions for such rare classes can be improved.

3 Analysis of the current situation

TFMs reach excellent overall performance on small datasets, but recent studies show that they tend to favour frequent classes: on imbalanced data, they often miss the minority class while remaining accurate overall [3]. On imbalanced clinical data from emergency and critical care, their advantage over classical methods also varies from one dataset to another [4]. First remedies have been proposed, such as adjusting the decision rule or changing the composition of the context [5, 6], and selecting the context examples closest to each query improves TFMs in general [7]. However, these works consider binary problems in which the minority class still has many examples. How TFMs behave when a class is represented by only a handful of samples within a multi-class problem, and which context should be given to the model in this case, remain open questions.

4 Scientific proposal

The student will study the behaviour of state-of-the-art TFMs on rare classes and propose strategies to improve it. The project will focus on:

- Evaluating the behaviour of TFMs on minority classes, from moderately imbalanced to very rare classes with only a few examples, in comparison with standard machine learning methods;

- Exploring strategies to mitigate this behaviour, starting from the choice of the context given to the model (which examples, how many and in which proportions), then considering adjustments of the predictions or a light adaptation of the model;
- Validating the approach on biomedical datasets with naturally rare classes, such as cardiac arrhythmias from electrocardiograms [8, 9], or breast cancer subtypes described by clinical data and images [10].

5 Expected results

The internship will provide an evaluation of the limitations of TFMs on long-tailed and rare classes, and propose a strategy, in particular for building the context, to improve their predictions on rare classes. Results are expected to clarify to what extent TFMs can be trusted for clinical categories with only a handful of patients. Potential extensions include multimodal settings combining clinical data with medical images [11] and the detection of classes that are absent from the context.

How to apply?

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- Documents to include in the application: resume, transcripts.

[1] Noah Hollmann et al. "Accurate predictions on small data with a tabular foundation model". In: *Nature* 637.8045 (2025), pp. 319–326. DOI: 10.1038/s41586-024-08328-6.

[2] Jingang Qu et al. "TabICL: A Tabular Foundation Model for In-Context Learning on Large Data". In: *Proceedings of the 42nd International Conference on Machine Learning*. Vol. 267. Proceedings of Machine Learning Research. PMLR, 2025, pp. 50817–50847. arXiv: 2502.05564.

[3] Junqi Cui et al. "Precision vs recall in surgical AI: performance and limitations of tabular foundation model for perioperative classification and risk prediction". In: *Digital Health* 12, 20552076261443585 (2026). DOI: 10.1177/20552076261443585.

[4] Yusuf Brima and Marcellin Atemkeng. *An Empirical Study of Machine Learning Robustness and Scalability for Imbalanced Tabular Clinical Data in Emergency and Critical Care*. 2025. arXiv: 2512.21602.

[5] Samuel McDowell, Nathan Stromberg, and Lalitha Sankar. "Correcting Class Imbalance in Prior-Data Fitted Networks for Tabular Classification". In: *ICML 2026 Workshop on Foundation Models for Structured Data*. 2026. arXiv: 2605.21742.

[6] Seunghan Lee. "Mitigating Label Shift in Tabular In-Context Learning via Test-Time Posterior Adjustment". In: *Proceedings of the 43rd International Conference on Machine Learning*. Vol. 306. Proceedings of Machine Learning Research. PMLR, 2026, pp. 64764–64788. arXiv: 2605.04363.

[7] Valentin Thomas et al. "Retrieval & Fine-Tuning for In-Context Tabular Models". In: *Advances in Neural Information Processing Systems*. Vol. 37. 2024, pp. 108439–108467. arXiv: 2406.05207.

[8] H. Guvenir et al. *Arrhythmia*. UCI Machine Learning Repository. 1997. DOI: 10.24432/C5BS32.

[9] Nils Strodthoff et al. "PTB-XL+, a comprehensive electrocardiographic feature dataset". In: *Scientific Data* 10, 279 (2023). DOI: 10.1038/s41597-023-02153-8.

[10] Lidia Garrucho et al. "A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations". In: *Scientific Data* 12, 453 (2025). DOI: 10.1038/s41597-025-04707-4. arXiv: 2406.13844.

[11] Wall Kim, Chaeyoung Song, and Hanul Kim. "MultiModalPFN: Extending Prior-Data Fitted Networks for Multimodal Tabular Learning". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026. arXiv: 2602.20223.